12. Recursive Descent Parser

CODE

```
#include <stdio.h>
#include <string.h>

char input[100];
int pos = 0;

int E();
int T();
int F();

int E()
{
    if (!T())
        return 0;

    while (input[pos] == '+')
    {
        pos++;

        if (!T())
            return 0;
    }

    return 1;
}
```

```
int T()
{
    if (!F())
        return 0;

    while (input[pos] == '*')
    {
        pos++;

        if (!F())
            return 0;
    }

    return 1;
}

int F()
{
    if (strncmp(&input[pos], "id", 2) == 0)
    {
        pos += 2;
        return 1;
    }

    if (input[pos] == '(')
    {
        pos++;
```

```
        if (!E())
            return 0;

        if (input[pos] == ')')
        {
            pos++;
            return 1;
        }

        return 0;
    }

    return 0;
}

int main()
{
    printf("Enter expression: ");
    scanf("%s", input);

    if (E() && input[pos] == '\0')
        printf("String is ACCEPTED.\n");
    else
        printf("String is REJECTED.\n");

    return 0;
}
```

OUTPUT

```
Enter expression: id+id*id
String is ACCEPTED.

--------------------------------
Process exited after 20.58 seconds with return value 0
Press any key to continue . . .
```